Ch 7.4: Cubic splines

Lecture 23 - CMSE 381

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Wed, Oct 29, 2025

Announcements

Last time:

- 7.2 Step functions
- 7.3 Basis functions

This lecture:

- 7.4 Cubic splines

Announcements:

- Homework # 6 is due 11/2!!
- Projects

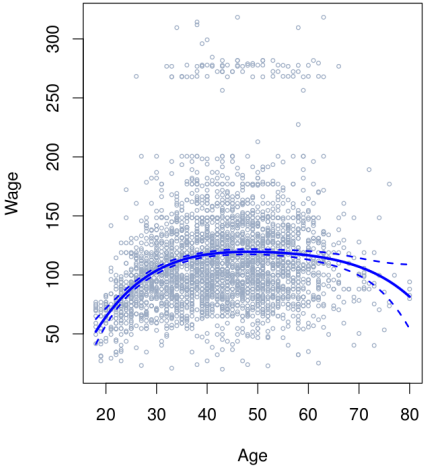
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	
24	F	10/31	Decision Trees	8.1	HW #6 Due Sun 11/2
25	M	11/3	Random Forests	8.2.1, 8.2.2	
26	W	11/5	Maximal Margin Classifier	9.1	
27	F	11/7	SVC	9.2	HW #7 Due Sun 11/9
28	M	11/10	SVM	9.3, 9.4	
29	W	11/12	Single Layer NN	10.1	
30	F	11/13	Multi Layer NN	10.2	HW #8 Due Sun 11/16
31	M	11/17	CNN	10.3	
32	W	11/19	Unsupervised learning / clustering	12.1, 12.4	
33	F	11/21	Virtual: Project Office Hours		HW #9 Due Sun 11/23
	M	11/24	***Review***		
	W	11/26	**Midterm #3**		
	F	11/28	Thanksgiving		
	M	12/1	Virtual: Project Office Hours		
	W	12/3	Virtual: Project Office Hours		
	F	12/5			**Project Due**

Section 1

Last time

Polynomial regression

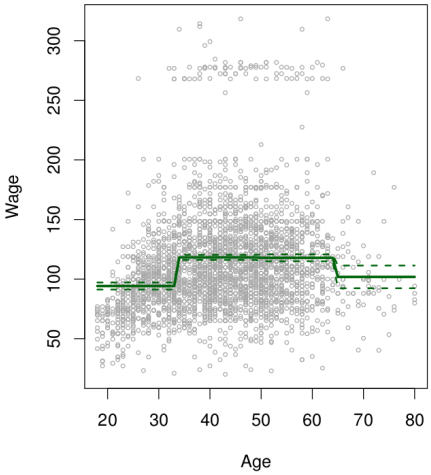
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$



Step function regression

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

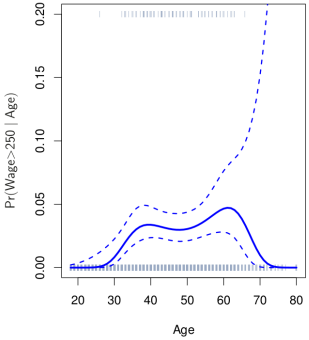


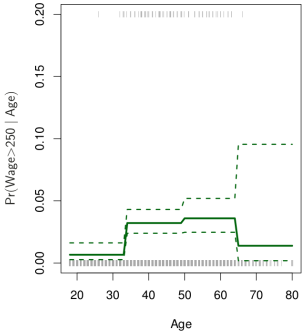
Classification version

$$\Pr(y_i > 250 \mid x_i) =$$

$$\frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$

$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$





Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

Section 2

Regression Splines

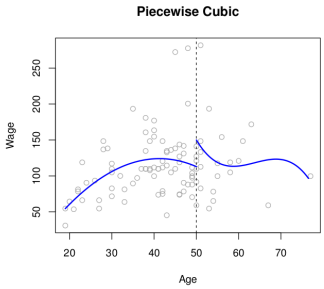
Piecewise polynomials

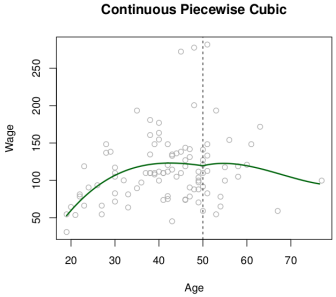
- Fit a polynomial regression

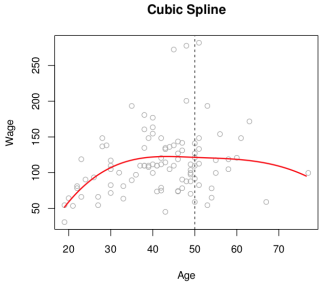
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

Building up to cubic splines



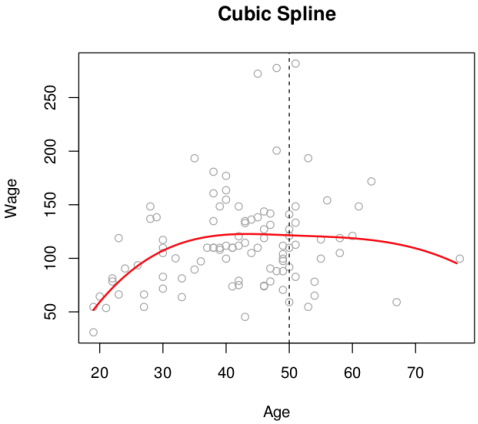




$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Cubic splines: degrees of freedom

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$



Spline basis representation

Want to pick b_i so that we represent a cubic spline with K knots as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Version 1: Truncated power basis function

$$h(x,z) = (x-z)_+^3 = \begin{cases} (x-z)^3 & \text{if } x > z \\ 0 & \textit{else} \end{cases}$$

Desmos link: https://www.desmos.com/calculator/ahllu5glar

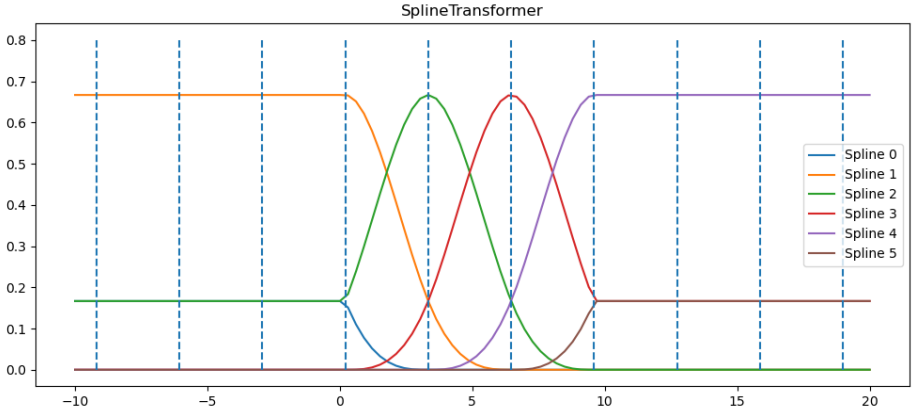
The (book) basis for cubic splines

Given knots at $z_1, \cdots, z_K$

- X
- X^2
- X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$

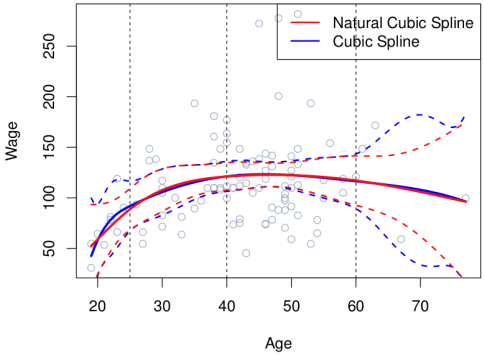
$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 h(X, z_1) + \beta_5 h(X, z_2) + \cdots + \beta_{k+3} h(X, z_K)$$

Version 2: B-spline basis function

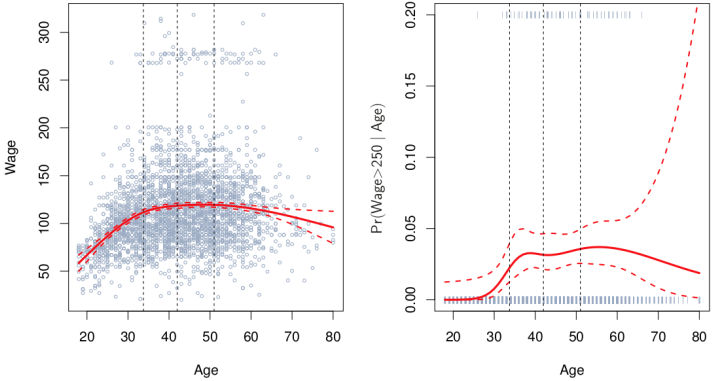


Coding example

Notes on cubic splines

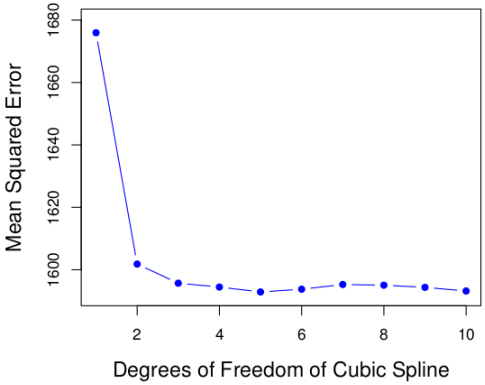


Where to put the knots?

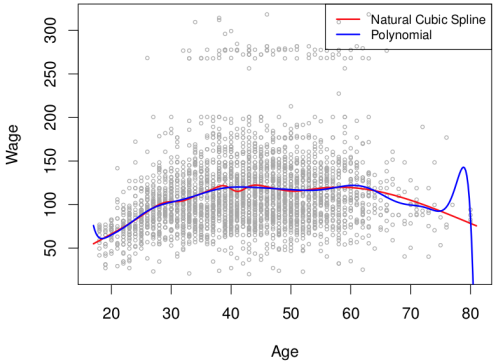


How many knots to use?

When in doubt, Cross-Validate.



Cubic splines vs Polynomial Regression



Next time

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21	W	3/19	Polynomial & Step Functions	7.1-7.2	
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24	W	3/26	Decision Trees	8.1	HW #6 Due Wed 3/26
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26	M	3/31	Maximal Margin Classifier	9.1	
27	W	4/2	SVC	9.2	
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	F	4/18	***Review***		
	M	4/21	**Midterm #3**		
	W	4/23			
	F	4/25			**Project Due**